

PATHNEO

Career Counselling & Career Direction Platform

Powered by AI. Built for Bharat.

Software Requirements Specification (SRS)
+
High Level Design (HLD)

Document Type	SRS + HLD Combined Document
Version	1.0 — Initial Release
Product Name	Pathneo
Target Market	India — 10th & 12th Grade Students
Prepared For	Founding Team / Investors / Development Partners

Pathneo Technologies Pvt. Ltd. | India

Part 1: Software Requirements Specification (SRS)

1.1 Purpose of This Document

This SRS document formally defines the functional and non-functional requirements for Pathneo — an AI-powered career counselling and career direction platform built specifically for Indian students of Class 10 and Class 12. This document is meant for the founding team, technical architects, product managers, potential investors, and development partners.

The goal of this document is to make sure everyone — from the developer sitting in Bangalore to the investor sitting in Mumbai — has a clear and shared understanding of what Pathneo will do, how it will behave, and what constraints it must operate within.

1.2 Product Vision

Pathneo aims to be the most trusted AI career companion for every Indian student — whether they are from a top school in Delhi or a government school in rural Vidarbha. We believe every student deserves personalised, intelligent, and affordable career guidance — not just those who can afford a ₹50,000 counselling package.

1.3 Scope

Pathneo will be delivered as:

- A mobile application (Android-first, then iOS)
- A web application (for school counsellors and parents)
- A WhatsApp-integrated notification and nudge layer
- An API layer for school/government integration

The platform will serve:

- Students (Primary Users) — Class 10 and Class 12, age 14 to 18
- Parents (Secondary Users) — who guide and finance education decisions
- School Counsellors (Tertiary Users) — who manage 100s of students
- School Administrators (B2B Buyers) — who purchase institutional licenses
- Government Bodies (B2G Buyers) — state education boards, CBSE, NEP implementation teams

1.4 Definitions and Abbreviations

Term	Full Form / Meaning
AI	Artificial Intelligence

Term	Full Form / Meaning
SRS	Software Requirements Specification
HLD	High Level Design
RIASEC	Holland's Career Interest Model (Realistic, Investigative, Artistic, Social, Enterprising, Conventional)
NEP 2020	National Education Policy 2020
NLP	Natural Language Processing
LLM	Large Language Model (e.g. GPT-4, Claude)
UDISE+	Unified District Information System for Education Plus (Govt of India)
B2B	Business to Business (selling to schools)
B2G	Business to Government (selling to state/central government)
Freemium	Free tier available, paid features unlocked on subscription
PWA	Progressive Web Application
API	Application Programming Interface

1.5 Target User Personas

Persona 1 — Aarav (The Confused Student)

- Age: 16, Class 11, Nagpur
- Scored 78% in Class 10, chose Science under family pressure
- Secretly loves sketching and wants to know if design is a real career
- Uses Instagram and YouTube daily, not comfortable with forms
- Needs: A friend to talk to, not a test to fill

Persona 2 — Sunita Bai (The Worried Parent)

- Age: 44, homemaker, Wardha district
- Husband is a farmer. First-generation college family.
- Hears only about IIT and MBBS from neighbours
- Needs: Simple explanation of career options in Marathi, ROI clarity, trust signals

Persona 3 — Mr. Ramesh Kulkarni (The Overloaded School Counsellor)

- Age: 38, counsellor at a government school, handles 600 students alone
- Has no digital tool. Uses Excel and notebooks.
- Needs: A dashboard that shows which students need attention and what to say to them

Persona 4 — District Education Officer (DEO)

- Age: 52, IAS or PCS officer, Nashik
- Wants data to present in district-level education meetings
- Needs: Aggregate career preference data, vocational uptake reports, UDISE-linked insights

1.6 Functional Requirements

FR-01 AI Career Counselling Module

- The system shall provide a conversational AI interface (text + voice) in English and supported Indian languages
- The AI shall conduct a natural-language career counselling session — not a rigid question form
- The AI shall ask open-ended questions about the student's interests, strengths, daily habits, fears, and aspirations
- The AI shall detect emotional signals (confusion, pressure, excitement) and adapt its tone accordingly
- Each session shall last 15 to 40 minutes depending on student engagement
- The system shall maintain session history and build a cumulative student profile across multiple sessions
- The AI shall never make a definitive career decision on behalf of the student — it shall present options with clarity

FR-02 Psychometric Assessment Engine

- The system shall administer an adaptive psychometric test based on RIASEC + aptitude + emotional intelligence models
- The assessment shall not look or feel like a school exam — it shall be gamified and visual
- Assessment results shall map to a ranked list of career clusters with match percentage
- The system shall re-run assessments every 6 months and show students how their profile has evolved
- Parent-facing summary of assessment shall be available in simplified language

FR-03 Career Path Generator

- For each recommended career, the system shall generate a personalised roadmap showing: required stream in Class 11, entrance exams, top colleges, typical salary progression, and future-proof score
- The system shall cover minimum 800 career paths including conventional (engineering, medicine, law) and emerging (UX design, sports science, AI ethics, content law, animation)
- Each career shall have a 'Reality Check' section showing actual day-in-the-life content

- The system shall show career options relevant to the student's board (CBSE / State Board) and location

FR-04 Stream Selection Module (Class 10 Specific)

- A dedicated 'Stream Fit' diagnostic shall be available for Class 10 students post-board result
- The module shall map the student's aptitude and interest to Science, Commerce, Arts, or Vocational stream
- The module shall show career outcomes for each stream with Indian context and salary data
- A 'Stream vs Dream' conflict resolution wizard shall be available when student preference conflicts with parent expectation

FR-05 Parent Dashboard

- Parents shall have a separate login with a simplified view of their child's career exploration
- The dashboard shall show: which careers the student is exploring, AI counsellor summary, next recommended action
- Parents shall receive weekly WhatsApp updates in their preferred language (Hindi, Marathi, Telugu, Tamil, Bengali, etc.)
- A 'Parent University' module shall educate parents about modern careers via short video explainers
- ROI calculator shall show estimated cost of education and expected salary return for any career path

FR-06 School Counsellor Co-Pilot Dashboard

- School counsellors shall have a web dashboard showing all students in their institution
- Dashboard shall highlight students who are: undecided, showing distress signals, or high-potential for non-conventional paths
- Counsellor shall receive AI-drafted talking points before each student session
- Counsellor shall be able to add manual notes that enrich the AI's understanding of a student
- Monthly automated reports shall be generated per school for principal/admin review

FR-07 College & Entrance Exam Mapping

- The system shall maintain a live database of top colleges mapped to career paths
- Entrance exam calendar (JEE, NEET, CLAT, NDA, NIFT, CMAT, CUET, etc.) shall be integrated
- Students shall be able to shortlist colleges and get an AI-generated preparation roadmap
- Cut-off data from previous 5 years shall be available per college per stream

FR-08 Multilingual Support

- The platform shall support 22 scheduled Indian languages as listed under the Indian Constitution
- AI counselling sessions shall be available in: Hindi, Marathi, Telugu, Tamil, Kannada, Bengali, Gujarati, and English at launch
- All UI text, notifications, and assessment content shall be translatable via CMS
- Voice input in regional languages shall be supported via speech-to-text integration

FR-09 Gamification and Engagement

- Students shall earn 'Clarity Points' for completing sessions, assessments, and exploring career modules
- Streak rewards shall be given for daily platform engagement
- A 'Career Explorer' mode shall allow students to casually browse careers like a social feed — not as a formal assessment
- Monthly 'Career Challenges' shall be available at district and state level

FR-10 Data & Analytics for Government

- Anonymised and aggregated career preference data shall be available as government-facing reports
- Reports shall be available at school, district, and state level
- UDISE+ integration shall be supported for automatic school data import
- NEP 2020 compliance reports (vocational exposure, multi-disciplinary interests) shall be auto-generated

1.7 Non-Functional Requirements

NFR ID	Category	Requirement
NFR-01	Performance	App shall load in under 3 seconds on a 4G connection with 2GB RAM Android device
NFR-02	Scalability	System shall handle 1 million concurrent users without degradation
NFR-03	Availability	Platform uptime shall be 99.9% — less than 8.7 hours downtime per year
NFR-04	Security	All student data shall be encrypted at rest (AES-256) and in transit (TLS 1.3)
NFR-05	Privacy	Platform shall comply with DPDPA 2023 (Digital Personal Data Protection Act, India)
NFR-06	Accessibility	Platform shall meet WCAG 2.1 Level AA accessibility standards
NFR-07	Offline Mode	Core career exploration modules shall function in offline mode with local data sync

NFR ID	Category	Requirement
NFR-08	Data Localisation	All student personal data shall be stored on servers located within India
NFR-09	AI Safety	AI responses shall be filtered for harmful, biased, or misleading career advice
NFR-10	Latency	AI counsellor response shall be generated within 4 seconds of user input

1.8 Constraints

- The product must work on Android phones with minimum 2GB RAM — this covers 80%+ of Indian student devices
- Data consumption per session should not exceed 15MB on 4G — many students are on limited data plans
- The platform must NOT require email registration — OTP-based mobile login is mandatory for Indian users
- AI counsellor must not replace a human counsellor — it is a co-pilot, not a replacement
- All career salary data must be India-specific — US salary data is misleading and counterproductive

1.9 Assumptions

- The student has access to a smartphone with internet connectivity at least 3 times per week
- Schools have at least one counsellor or teacher who can serve as the Pathneo admin
- Government integration (UDISE+) will require a separate API partnership agreement
- AI model (LLM) will be accessed via third-party API (e.g. Anthropic Claude or OpenAI GPT) initially, with fine-tuning planned for Year 2

Part 2: High Level Design (HLD)

2.1 System Architecture Overview

Pathneo is designed as a cloud-native, microservices-based platform hosted on Indian data centre infrastructure. The architecture follows a layered design — Client Layer, API Gateway, Service Mesh, Data Layer, and AI Engine — ensuring scalability, modularity, and maintainability.

Architecture Style: Microservices on Kubernetes | Cloud: AWS Mumbai Region (ap-south-1) | Database: PostgreSQL (primary) + MongoDB (AI sessions) + Redis (cache) | AI Engine: LLM via API (Anthropic/OpenAI) with RAG layer

2.2 High Level Architecture Layers

Layer 1 — Client Applications

Client	Technology	Primary Users	Key Features
Android App	React Native + Native Modules	Students	AI counselling, assessments, career explorer, gamification
iOS App	React Native (Phase 2)	Students (premium)	Same as Android app
Web App (PWA)	Next.js + PWA	Parents, Counsellors	Dashboard, reports, WhatsApp integration
Admin Portal	React.js	School Admins, Govt	Analytics, bulk uploads, licence management
WhatsApp Bot	WhatsApp Business API	Parents	Weekly nudges, career tips, alerts in regional language

Layer 2 — API Gateway & BFF Layer

- API Gateway: AWS API Gateway / Kong for routing, rate limiting, authentication
- BFF (Backend For Frontend): Separate BFF services for Mobile, Web, and Admin — each optimised for its client
- Authentication: OTP-based login via Firebase Auth + JWT token issuance
- Rate Limiting: 100 requests per minute per user to prevent AI cost abuse
- CDN: CloudFront for static assets, media files, and offline-capable content bundles

Layer 3 — Core Microservices

Service Name	Responsibility	Technology Stack
User Service	Registration, login, profile, roles (student/parent/counsellor/admin)	Python + PostgreSQL
Assessment Service	Psychometric test delivery, scoring, RIASEC mapping	Python + PostgreSQL
AI Counsellor Service	LLM session management, conversation history, tone adaptation	Python + LangChain + MongoDB
Career Engine Service	Career database, path generation, college mapping, entrance exam data	Python + PostgreSQL
Notification Service	WhatsApp, push notifications, SMS, email delivery	Python + AWS SNS + WhatsApp API
Analytics Service	Student engagement, school dashboards, government reports	Python + ClickHouse
School Admin Service	Licence management, student bulk onboarding, counsellor tools	Python + PostgreSQL
Payment Service	Subscription management, school licence billing, refunds	Python + Razorpay
Content Service	Career content CMS, multilingual content, video metadata	Python + MongoDB + S3
Gamification Service	Points, streaks, challenges, leaderboards	Python + Redis

Layer 4 — AI Engine

- Primary LLM: Anthropic Claude API (for safety, nuance, and instruction-following) or OpenAI GPT-4o
- RAG Layer: Career knowledge base stored in Pinecone vector database, retrieved at runtime to ground AI responses with Indian-specific data
- Prompt Engineering: Role-specific system prompts for 'Empathetic AI Counsellor' persona — strict guardrails against harmful or biased career advice
- Speech-to-Text: Google Cloud Speech API for voice input in Indian languages
- Text-to-Speech: ElevenLabs or Azure TTS for warm, human-like AI voice output
- AI Safety Layer: Custom classifier running before every AI response to check for harmful content, incorrect salary data, or medical/legal advice
- Fine-Tuning Plan (Year 2): Fine-tune a base model on Indian career counselling conversations to reduce LLM API cost by 60%

Layer 5 — Data Layer

Database	Type	What It Stores	Why This Choice
PostgreSQL (RDS)	Relational	Users, assessments, schools, colleges, career data	Strong ACID compliance for financial and profile data

Database	Type	What It Stores	Why This Choice
MongoDB (Atlas)	Document	AI session transcripts, conversation history	Flexible schema for variable-length AI conversations
Redis (ElastiCache)	Cache / Queue	Session tokens, rate limits, gamification state, real-time leaderboards	Sub-millisecond reads for real-time features
ClickHouse	Analytical	Event logs, engagement analytics, government reports	Columnar DB — extremely fast for aggregation queries
Pinecone	Vector	Career knowledge embeddings for RAG retrieval	Purpose-built for semantic search, fast at scale
S3 (AWS)	Object Storage	Career videos, documents, report PDFs, content assets	Durable, cheap, CDN-compatible

2.3 Data Flow — AI Counselling Session

Below is the step-by-step data flow when a student starts an AI counselling session:

1. Student opens app and taps 'Talk to Pathneo AI'
2. Client app sends session start request to API Gateway with JWT token
3. API Gateway routes to AI Counsellor Service; User Service confirms identity and fetches student profile
4. AI Counsellor Service loads: student profile, previous session summaries, RIASEC scores, and relevant career context chunks from Pinecone vector DB
5. System prompt is assembled: persona + student context + retrieved career knowledge
6. Student sends first message (text or voice); voice is transcribed via Google STT
7. LLM API is called with assembled context + student message
8. Response passes through AI Safety Classifier — if flagged, fallback response is returned
9. Response is sent to client; simultaneously stored in MongoDB session store
10. At session end, a session summary is generated and saved to student profile in PostgreSQL
11. Notification Service triggers a post-session WhatsApp message to parent with session summary

2.4 Security Architecture

- Authentication: OTP via SMS (MSG91) + JWT + Refresh Token rotation
- Authorisation: Role-based access control (RBAC) — Student / Parent / Counsellor / Admin / Govt roles
- Data Encryption: AES-256 at rest, TLS 1.3 in transit, field-level encryption for sensitive PII
- DPDPA 2023 Compliance: Consent management module, data deletion workflow, grievance officer designation

- AI Data Isolation: Student conversation data is never used to train external LLMs without explicit opt-in consent
- Penetration Testing: Quarterly third-party pen testing by a CERT-In empanelled firm
- WAF: AWS WAF + Shield Standard for DDoS protection

2.5 Scalability Design

- Horizontal Scaling: All microservices run in Kubernetes pods with auto-scaling based on CPU and memory thresholds
- LLM Cost Control: Caching of common AI responses, streaming responses to reduce latency perception, queue-based processing during peak load
- Database Scaling: PostgreSQL read replicas for analytics queries; MongoDB sharding for session data
- Peak Load Events: Board exam result season (May-June) is the highest traffic period — auto-scale groups pre-warmed 48 hours before result dates
- CDN Strategy: Career content, assessment media, and onboarding videos served via CloudFront edge locations in India

2.6 Integration Architecture

Integration	Partner / API	Purpose
LLM (AI Brain)	Anthropic Claude / OpenAI GPT-4o	Core AI counselling and content generation
Speech Recognition	Google Cloud Speech-to-Text	Voice input in Indian languages
Text to Speech	Azure TTS / ElevenLabs	AI voice output for counselling sessions
WhatsApp Notifications	WhatsApp Business API (via Gupshup or Interakt)	Parent nudges, session summaries, exam reminders
SMS / OTP	MSG91 or Kaleyra	OTP-based login, SMS alerts
Payment Gateway	Razorpay	Student subscriptions, school licence billing
Vector Database	Pinecone	RAG-based career knowledge retrieval
Analytics & Logging	AWS CloudWatch + Datadog	System monitoring, error alerting, cost tracking
Government Data	UDISE+ API (NIEPA)	School data import for government-linked schools
Email	AWS SES	Transactional emails for school admins

2.7 Deployment Architecture

- Cloud Provider: AWS ap-south-1 (Mumbai) as primary region

- Disaster Recovery: AWS ap-south-2 (Hyderabad) as DR region with RTO of 4 hours, RPO of 1 hour
- Container Orchestration: Kubernetes (EKS) with Helm charts for each microservice
- CI/CD: GitHub Actions for automated testing; ArgoCD for GitOps-based deployment to Kubernetes
- Environment Strategy: Development, Staging (with anonymised production data), and Production — each isolated
- Infrastructure as Code: Terraform used for all AWS resource provisioning — no manual console changes in production
- Monitoring: Prometheus + Grafana dashboards; PagerDuty for on-call alerting

2.8 Monetisation Architecture

Revenue Stream	Model	Pricing Indicative	Tech Implication
Student Freemium	Free + Paid Subscription	Free: 3 sessions/month; Pro: ₹499/month	Feature flags per subscription tier
School B2B License	Annual Per-Student Fee	₹15 to ₹50 per student per year	Multi-tenant school portal, bulk invoicing via Razorpay
Government B2G	Annual Platform License	Custom pricing per state	UDISE+ integration, custom report generation module
Career Content Premium	One-time purchase	₹199 to ₹499 per career deep-dive	Content DRM, offline access with time-limited license
College Partner Listings	B2B partnership	₹50,000 to ₹5L per year per college	College admin portal, application tracking module

2.9 Development Phasing

Phase 1 — MVP (Months 1 to 4)

- AI counselling session (text only, English + Hindi)
- Basic RIASEC psychometric assessment
- Career path generator for top 100 careers
- Student mobile app (Android)
- Basic parent dashboard (web)
- OTP login, student profiles, session history

Phase 2 — Growth (Months 5 to 9)

- Voice input support in 4 Indian languages
- Stream selection module for Class 10 students
- School counsellor dashboard (web)

- WhatsApp parent notification integration
- Gamification — Clarity Points and streaks
- College and entrance exam mapping module
- Android + iOS app (React Native)

Phase 3 — Scale (Months 10 to 18)

- Full 22-language support
- Government reporting module (UDISE+ integration)
- Parent University content module
- B2G pilot with 2 state governments
- LLM fine-tuning on Indian career data
- District-level talent mapping analytics
- National Career Olympiad feature

2.10 Tech Stack Summary

Category	Technology Chosen	Rationale
Mobile App	React Native	Single codebase for Android and iOS, large Indian developer community
Web Frontend	Next.js (React)	SSR for SEO, fast page loads, PWA support
Backend Services	Python + Python	Node for APIs, Python for AI/ML workloads
Primary Database	PostgreSQL on AWS RDS	Reliable, ACID-compliant, great for relational data
Session / AI Data	MongoDB Atlas	Flexible schema, ideal for conversation transcripts
Cache + Queue	Redis (AWS ElastiCache)	Real-time features, session management, job queues
Analytics DB	ClickHouse	10x faster than PostgreSQL for aggregation, perfect for govt reports
AI / LLM	Anthropic Claude API	Superior instruction-following, strong safety guardrails
Vector Search	Pinecone	Managed, scalable, fast semantic retrieval for RAG
Infrastructure	AWS (Mumbai Region)	Data localisation compliance, best coverage in India
Container Mgmt	Kubernetes (EKS)	Industry standard, excellent auto-scaling
CI/CD	GitHub Actions + ArgoCD	GitOps workflow, fast deployments
Payments	Razorpay	Best Indian payment gateway, supports UPI, EMI, invoicing

Category	Technology Chosen	Rationale
WhatsApp	Gupshup / Interakt	Reliable WhatsApp Business API partners in India

Appendix — Key Metrics for Success

Metric	Target (End of Year 1)	Measurement Method
Registered Students	5,00,000	User Service DB
Monthly Active Users	1,50,000	Analytics Service
Average Sessions per Student per Month	4 sessions	Session logs
Parent Dashboard Activation Rate	40% of student accounts	User Service
School Partnerships (B2B)	200 schools	Admin portal
Career Paths Covered	800+	Content Service DB
Net Promoter Score (NPS)	60+	In-app surveys
AI Session Satisfaction Rating	4.2 out of 5	Post-session rating
Languages Supported at Launch	8	CMS language config
Data Uptime SLA	99.9%	CloudWatch

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